JERY RODRÍGUEZ

SOFTWARE & AI ENGINEER
COMPUTER SCIENCE STUDENT
& MATH LOVER

CONTACT

Havana, Cuba — Remote
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SKILLS

LANGUAGES

Python, C, C++, C#, Java, JavaScript, TypeScript, Haskell, Prolog & Rust.

FRAMEWORKS & TOOLS

React, Node.js, .NET, Unity, PyTorch, Git, GitLab, GitHub, Linux, FastAPI, Docker, Scikit-learn, Kubernetes, Pandas, NumPy, Matplotlib, Seaborn, LaTeX, PyCharm, Photoshop, Illustrator & VS Code.

SPECIALIZATIONS

- Full-Stack Development
- Applied Mathematics
- UI Design
- Game development
- Compiler Design
- Mathematical Modeling & Optimization

LANGUAGES

- Spanish - Native
- English - B2 (advancing toward C1)

EXECUTIVE SUMMARY

I enjoy working with applied mathematics and bringing that analytical mindset into computational development. I work closely with teams to ship scalable applications, strengthen backend systems, and craft visual designs that elevate the overall user experience. Right now I'm deepening my work in artificial-intelligence architectures and advanced neural-network design, focusing on their optimization, training, and deployment in real-world environments.

CORE EXPERIENCE

AI RESEARCH & DEVELOPMENT — MATCOM, UH

- Developed an autonomous Hex-playing agent that mixes Minimax with alpha–beta pruning and Monte Carlo Tree Search, designing a multi-signal heuristic with a Genetic Algorithm that tunes its evaluation weights through competitive self-play.
- Contributed to the validation of CecilA (Cuban National AI) training datasets.
- Designed and Built a dynamic neural network with adaptive retraining capabilities, enabling continuous learning and iterative model refinement.

BACKEND DEVELOPMENT — MATCOM, UH & INDEPENDENT

- Developed web applications in collaboration with cross-functional teams, specializing in business logic and database development using TypeORM with MySQL.

FRONTEND DEVELOPMENT — INDEPENDENT & DEV COMPANIES

- Built responsive interfaces for several businesses in collaboration with development teams.
- Designed and built UI interfaces for desktop applications on both Windows and Linux platforms, embedding core business logic.

COMPUTER ARCHITECTURE — MATCOM, UH

- Designed and built a functional 32-bit microprocessor architecture in Logisim with an integrated cache system, manipulating low-level language: assembler.

APPLIED MATHEMATICS — INDEPENDENT

- Redefined classical optimization methods — Newton's Method and Steepest Descent — specifically for quadratic models.
- Worked with numerical methods, addressing precision issues and managing numerical errors.
- Worked on simulation problems, developing a discrete-event population model with probabilistic life events and demographic dynamics.
- Developed statistical predictive models to analyze academic dropout and success, combining descriptive analysis, visualization, clustering, and classification techniques.
- Performed exploratory and predictive analysis on social media data, identifying patterns, engagement trends

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- Designed and implemented a vector-space model applied to psychology, enabling the comparison of individuals by quantifying how similar their personalities are across multiple psychological variables.
- Designed and implemented a vector-space search model based on the Generalized Vector Space Model (GVSM), supported by lightweight indexing structures (Patricia Trie and co-occurrence index).

GAME DEVELOPMENT — MATCOM, UH

- Developed a game in Unity and built it for Windows. It's a Strategy card game inspired by the Game of Thrones universe, featuring a custom interpreter that transforms code into playable.

DESIGN VISUAL — INDEPENDENT

- Created numerous visual designs, logos, and prototypes for businesses and organizations across different industries. Specialized in Photoshop, delivering professional and versatile results in design projects.

EDUCATION

B.S. IN COMPUTER SCIENCE — University of Havana (MATCOM)

Academic Focus: Artificial Intelligence, Data Structures, Algorithms, Cryptography, Distributed Systems, Compilers, Software Engineering, Simulation, Applied Mathematics, Databases & Computer Architecture.

DISTINCTION: Software Engineering, Applied Mathematics, Data Structures & Algorithms

AVAILABILITY & RELOCATION

IMMEDIATE: Available for Full-Stack and other technical roles

FUTURE: Open to global relocation

MISSION: I aim to keep learning and expanding my abilities every day. I strive to collaborate with multidisciplinary teams to build meaningful and impactful solutions. I am committed to creating and innovating, turning ideas into real products that make a difference.